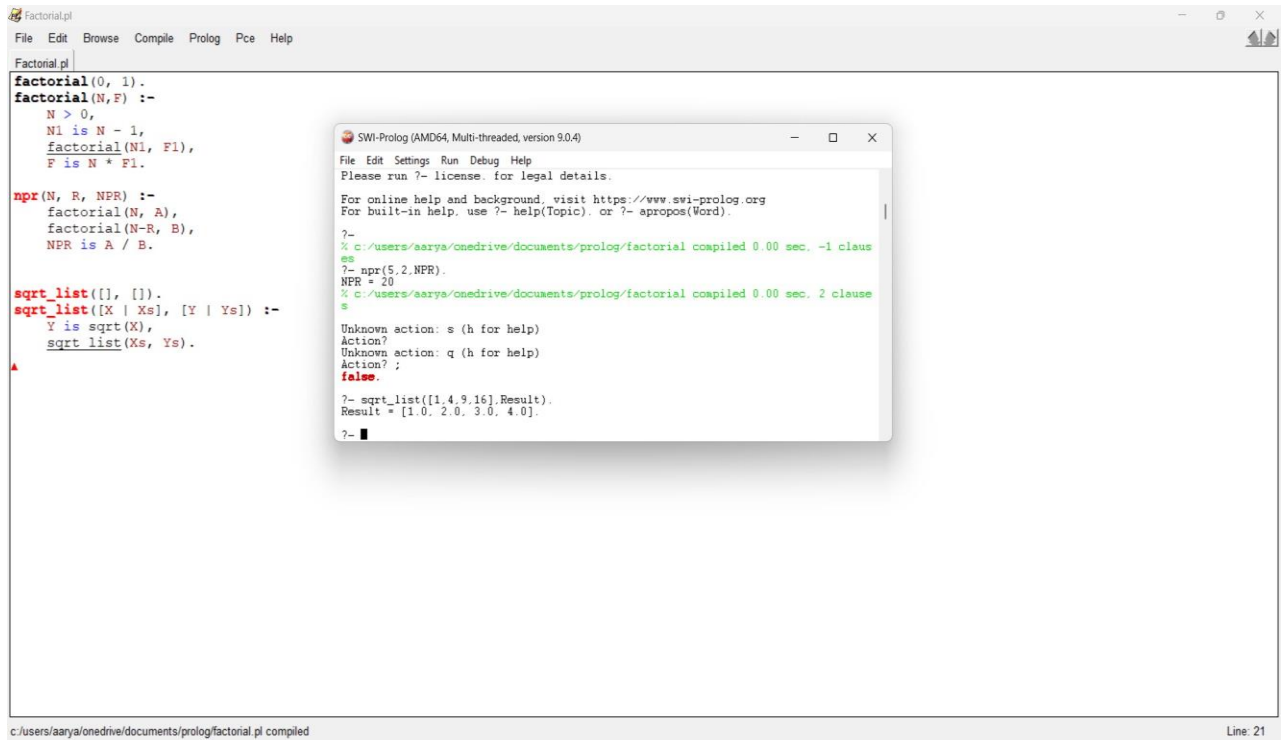


Aarya Arban

S11-07

Aim: To implement functions in Prolog



The screenshot shows a Prolog IDE window titled 'Factorial.pl' with a menu bar (File, Edit, Browse, Compile, Prolog, Pce, Help). The main editor contains the following Prolog code:

```
factorial(0, 1).
factorial(N, F) :-
    N > 0,
    N1 is N - 1,
    factorial(N1, F1),
    F is N * F1.

npr(N, R, NPR) :-
    factorial(N, A),
    factorial(N-R, B),
    NPR is A / B.

sqrt_list([], []).
sqrt_list([X | Xs], [Y | Ys]) :-
    Y is sqrt(X),
    sqrt_list(Xs, Ys).
```

Below the editor, a console window titled 'SWI-Prolog (AMD64, Multi-threaded, version 9.0.4)' displays the execution results:

```
File Edit Settings Run Debug Help
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/users/aarya/onedrive/documents/prolog/factorial compiled 0.00 sec. -1 clause
ss
?- npr(5.2, NPR).
NPR = 20
% c:/users/aarya/onedrive/documents/prolog/factorial compiled 0.00 sec. 2 clause
s
Unknown action: s (h for help)
Action?
Unknown action: q (h for help)
Action? ;
false.

?- sqrt_list([1,4,9,16], Result).
Result = [1.0, 2.0, 3.0, 4.0].

?-
```

The status bar at the bottom indicates 'c:/users/aarya/onedrive/documents/prolog/factorial.pl compiled' on the left and 'Line: 21' on the right.